

Seeing Sound: The Role of GUI Aesthetics in AI Audio Generation

Anish Ghosh, *UM Graduate Student*
(axg1652@miami.edu)

University of Miami, Miami, USA

The advancement of AI sound generation technologies have a distinct impact on the shaping of future tools for music and audio creation. Specifically, graphical user interfaces (GUIs) can influence how these tools are perceived as they become more accessible. This study addresses the gap in understanding how different AI audio generation models and graphical user interfaces (GUIs) influence user perceptions of audio quality and usability, particularly in the context of music and audio creation tools. This study explores the intersection of artificial intelligence (AI) audio generation and human-computer interaction (HCI), emphasizing the influence of graphical user interfaces (GUIs) on user perception. Using a 3x2 within-subjects factorial design, we investigated how different AI text-to-audio generators (AudioGen, CTAG, and AudioLDM) and GUIs impact participants' evaluations of audio accuracy, fidelity, and artistic interpretation. Twenty-one music engineering students from the University of Miami participated, completing an online survey featuring audio stimuli and GUI screenshots. Quantitative analyses revealed significant main effects for AI models on all metrics, with AudioGen outperforming in accuracy and artistic interpretation, and CTAG leading in fidelity. Although GUI differences did not yield statistically significant results, qualitative feedback indicated that more advanced GUIs enhanced perceived audio quality and usability. This study underscores the importance of thoughtful interface design in AI tools, with implications for fostering collaborative creativity. Future work should broaden participant diversity, refine GUI designs, and explore multimodal stimuli for a deeper understanding of HCI in AI audio applications.

0 INTRODUCTION

The convergence of artificial intelligence (AI) and sound generation has unlocked new opportunities for creativity and interaction in various domains, including music production, sound design, and immersive media. AI-driven systems like AudioGen and AudioLDM are transforming how sound is created, allowing users to generate complex audio outputs from simple text prompts. As these technologies evolve, the role of human-computer interaction (HCI) becomes increasingly significant. The interface through which users engage with AI systems not only influences their experience but may also shape their perceptions of the AI's creative output.

In this study, we focus on the relationship between AI audio generation and user interface design, specifically examining how different graphical user interfaces (GUIs) impact user evaluations of AI-generated audio. By integrating both qualitative and quantitative methodologies, we assess user ratings of audio accuracy, fidelity, and artistic interpretation across three distinct AI audio models. Additionally, this research aims to uncover the subtle yet powerful ways in which interface design alters user engagement and per-

ception, paving the way for more intuitive and impactful AI sound generation tools. This work contributes to the growing dialogue on human-AI collaboration, offering insights into how interface design can enhance creative processes and user satisfaction in AI-driven sound generation.

1 LITERATURE REVIEW

The integration of artificial intelligence (AI) into audio generation represents a significant advancement in both technology and its potential creative applications. This literature review dives into the relationship between AI sound generation systems and human-computer interaction (HCI), emphasizing user experience, interface design, and collaborative creativity [1, 2].

1.1 Advances in AI Audio Generation

Recent studies have highlighted various approaches to AI-driven audio generation including the development of text-to-audio models that allows users to input text and get an output of audio [3, 4]. Kwon and Lee (2023) propose a user-centered design framework for AI audio generation systems, emphasizing the need to enhance user experience

through intuitive interfaces [5]. Their model addresses the challenges users face when interacting with AI tools, providing insights into how the design of an interface can influence creative expression.

Cherep, Singh, and Shand (2024) present a novel approach to text-to-audio generation through synthesizer programming [6]. Their work leverages advanced AI models to generate sound, that push the boundaries of traditional music production. This paper specifically discusses creating a sound designing synthesizer to be used by novices and professionals. Similarly, Ghosal et al. (2024) introduce an instruction-guided latent diffusion model for text-to-audio generation, emphasizing the importance of user input in the creative process [7]. These methods not only enhance the quality of AI-generated sound design but also promote a more interactive and engaging user experience.

Ranjan, Tiwari, and Kumar (2022) conducted a systematic review of AI-based audio generation, highlighting the evolution from basic algorithmic composition to sophisticated generative models [8]. They suggest that these advancements necessitate a better understanding of user interaction, as effective AI tools must accommodate diverse creative needs and preferences.

1.2 Human-AI Collaboration in Audio Creation

The notion of collaboration between humans and AI is central to contemporary audio generation practices [9]. understanding user behavior in designing generative AI systems is also an important focus. Insights into user preferences can lead to more responsive and adaptable interfaces, ultimately improving the quality of musical output [10]

It has been posited technologies that enable exploration of the collaborative relationship between sound and AI can unlock new forms of creativity [11]. This work highlights how AI can serve not just as a tool but as a creative partner, reshaping the roles of human composers and musicians. This partnership is further supported by experimenters who examine how collaborative AI can enhance the creative process through intelligent interfaces, allowing users to experiment and innovate in ways that were previously unattainable [12]. There are even models of music composition computing based on Human-Computer Interaction [13, 14].

1.3 User Experience and Interface Design

The physiological and psychological effects of interactive musical interfaces have a positive correlation with stress and anxiety reduction, demonstrating the potential therapeutic benefits of music technology [15]. This research suggests that thoughtful interface design can significantly enhance user well-being and overall experience.

Shah and Trivedi (2023) delve into dialogue-based interactions in musical contexts, arguing that facilitating ongoing communication between users and AI enhances musical interaction [16]. This aligns with findings that illustrate how users can leverage AI to explore and expand their musical ideas [17]. Some studies go further to explain the role of novice AI systems in audio creation, emphasizing the

importance of user-friendly interfaces that cater to varying levels of expertise [18].

1.4 Perception of AI-Generated Sound

Understanding user perceptions of AI-generated audio is essential for the ongoing development of these technologies. Some studies investigate how individuals perceive music created by AI, revealing that familiarity with the technology influences user acceptance and enjoyment. This finding highlights the importance of educating users about AI capabilities to enhance their interaction with audio generation tools [19].

In conclusion, the development of AI audio technologies is linked with human-computer interaction principles. As AI systems become more sophisticated, developing meaningful user interaction and understanding diverse cultural contexts will be crucial for their success. Future research should continue to explore these dynamics, ensuring that AI audio generation becomes more collaborative and enriching than ever before.

1.5 Related Works

My paper is most similar to the user study developed by Cherep, Singh, and Shand (2024). They develop a user study in which the participants are asked to qualitatively and quantitatively analyze audio. They develop a study around playing audios and showing descriptions to participants and then having them rate accuracy, confidence, and artistic interpretation. I will use a similar study design with three metrics to but with an added visual element to test the effect of user interfaces on the perception of AI generated sounds [6]. Additionally, I will use the model created by Newman, Morris, and Lee (2023) to develop a user interface design to display to participants [20].

2 METHODS

In the methods section I will discuss all elements regarding an experiment to test the effects of different user interfaces on the perception of different models of AI text-to-audio generators.

2.1 Study Design

Our independent variables are types of text-audio-generator with 3 levels (LAILION-CLAP, AudioGen, and AUDIO-LDM) [6] and types of graphic user interface (GUI) with 2 levels. Our dependent variables are user ratings of accuracy, fidelity, and artistic interpretation of the sound effect stimuli that emerge from the prompt "sirens and a humming engine approach and pass." This is mostly a qualitative study that will ask for rankings with one quantitative question at the end of the survey. The study will be an experimental procedure that takes place during an online forum.

2.2 Participants

21 individuals participated in this study. They were all music engineering students in the Frost School of Music at

the University of Miami. We recruited participants through an online forum that helped students find participants for various studies. The average age of the participants was 23.

2.3 Stimuli (Materials)

Two different GUIs were designed to present AI-generated audio files. The first GUI focused on simplicity,

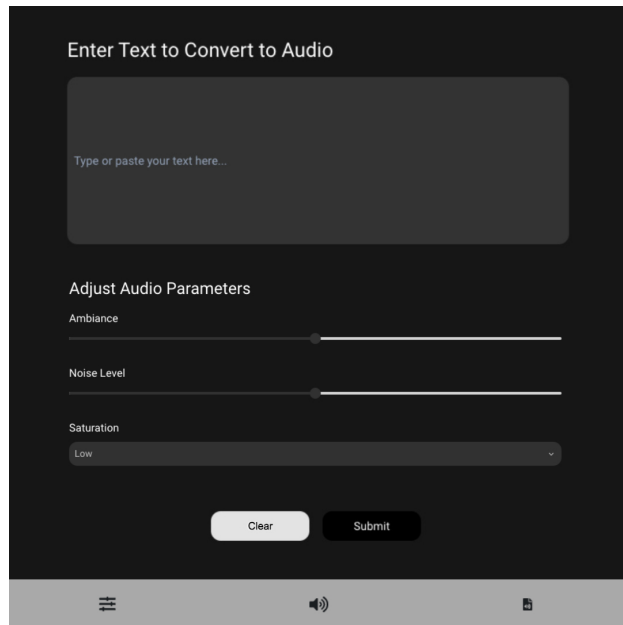


Fig. 1. This is an example of the GUI designed to have more parameters which will be labeled as UI2

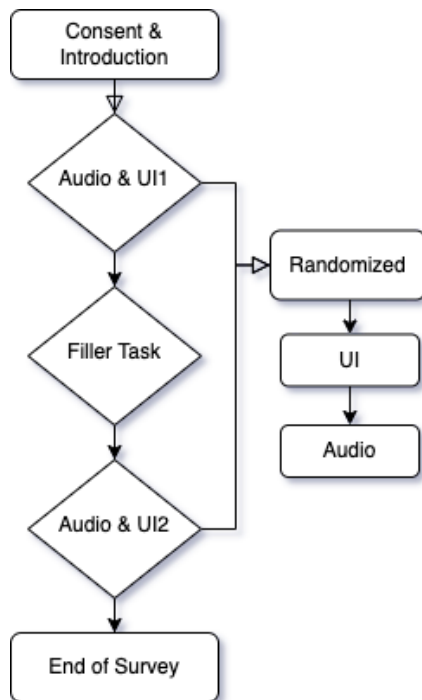


Fig. 2. This is the survey flow which was part of the conditions that participants undertook with randomization and counterbalanced conditions to mitigate order and fatigue effects.

while the second incorporated advanced features such as customizable audio parameters, volume, and version history as seen in Fig 1. Both UIs presented the same set of audio files generated from the text prompt "sirens and a humming engine approach and pass" by the AI model. In this experiment, we used 3 audio samples from LAILION-CLAP, Audio Gen, and Audio-LDM. We published a survey through the platform Qualtrics which had a total of 36 questions, including demographic questions, during questions, and post-questions. We asked them to rate each audio file based on accuracy, fidelity, and their individual artistic interpretation. Additionally, we asked participants to rate these same metrics while being provided a screenshot of the two GUIs for the models. Accuracy was measured as percentage and the other two ratings were based on a Likert scale.

2.4 Procedure

During an online forum for student survey projects in the Frost School of Music at University of Miami, we distributed a Qualtrics link via zoom and asked participants to sign the initial consent and participate in the survey. Students had about 20 minutes to complete the survey on their personal devices. The Survey began with a consent form and then an explanation of the instructions. After the first page, participants were asked a short set of demographic questions such as age, race, gender, and academic level. Then, the during questions began where participants listened to audio files and viewed screenshots of GUIs in random order and answered questions based on each page: This section was limited to six pages and contained a filler task in the middle to reduce chances of fatigue as seen in Fig 2. Finally, participants came to the post-questions and then a thank you page after the survey was completed. This study was a 3 x 2 within-subjects factorial design. Quantitative data from the surveys were analyzed using statistical methods, including ANOVA, to determine if there were significant differences in user experience ratings between the two interfaces. Qualitative responses were coded for thematic analysis, identifying common trends and insights related to user preferences and experiences.

2.5 Apparatus

This survey was administered online but most students took it on campus or near campus on their personal devices, either a phone or computer. They were advised to find a quiet place or bring headphones before the forum began, knowing they will possibly take surveys relating to music studies.

3 RESULTS

Analyzing the results from the study shows to what extent how GUI design influences the perception of AI-generated audio across three dependent variables: accuracy, fidelity, and artistic interpretation.

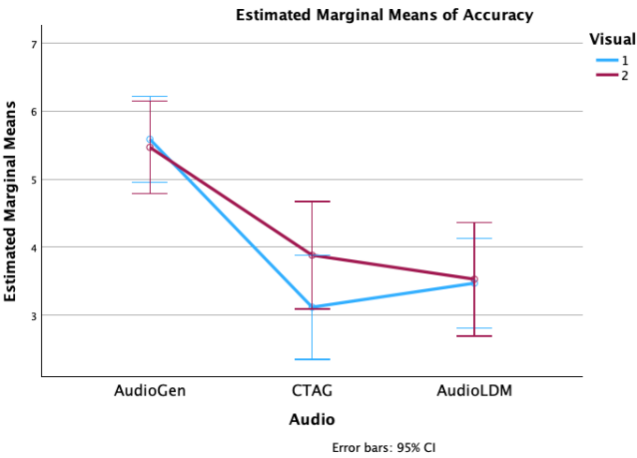


Fig. 3. Estimated Marginal Means of Accuracy for different AI audio generation models (AudioGen, CTAG, and AudioLDM) under two visual interface conditions (Visual 1 and Visual 2). The graph illustrates participants’ accuracy ratings with 95 percent confidence intervals. The lines represent the interaction between the audio models and the visual interfaces, highlighting the varying effects of the interfaces on perceived accuracy.

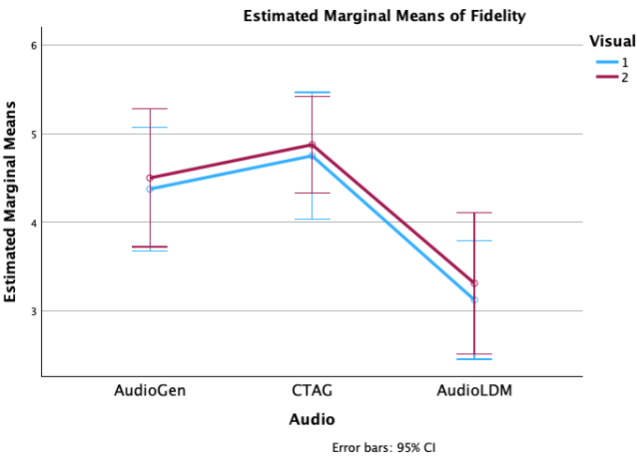


Fig. 4. Estimated Marginal Means of Fidelity for different AI audio generation models (AudioGen, CTAG, and AudioLDM) under two visual interface conditions (Visual 1 and Visual 2). The graph depicts participants’ fidelity ratings with 95 percent confidence intervals. The interaction between audio models and visual interfaces shows consistent patterns across conditions, with CTAG receiving the highest fidelity ratings overall.

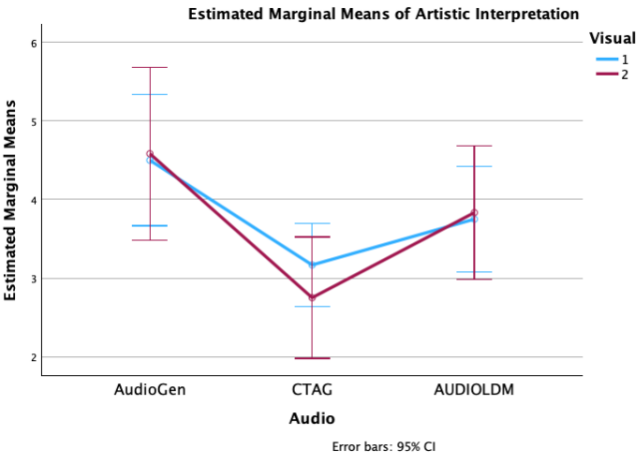


Fig. 5. Estimated Marginal Means of Artistic Interpretation for different AI audio generation models (AudioGen, CTAG, and Audi-oLDM) under two visual interface conditions (Visual 1 and Visual 2). The graph highlights participants’ artistic interpretation ratings with 95 percent confidence intervals. The results show variation across models, with AudioGen receiving higher artistic interpretation ratings, particularly under Visual 2, while CTAG shows the lowest ratings.

3.1 Quantitative Results

Accuracy ratings of the audio files in relation to the prompt "Sirens and a humming engine approach and pass" were measured on a 7-point Likert Scale. A two-way ANOVA revealed no significant interaction effect between the AI text-to-audio generator and the UI design on accuracy ratings with F-values ($F(2,32) = 2.172, p = .130$) but did reveal a significant main effect of audio model with F-values ($F(2,32) = 21.243, p < .001$). AudioGen performed the best in terms of accuracy ratings. See Fig. 3 for the condition means plot.

Fidelity ratings, measured on a 7-point Likert scale, showed a main effect for audio model ($F(2,30) = 11.129, p < .001$), with CTAG receiving the highest fidelity ratings across all audio generators. Additionally, AudioLDM produced the lowest fidelity ratings regardless of the UI used. See Fig. 4 for the condition means plot.

Artistic interpretation ratings, also based on a 7-point Likert scale, demonstrated no significant interaction effects between the UI design and AI models with F-values ($F(2,22) = .623, p = .546$). However, there was a main effect for AI audio model, with AudioGen receiving the highest artistic interpretation ratings overall ($F(2,22) = 6.822, p < .005$). See Fig. 5 for the condition means plot.

3.2 Follow Up Analysis

After finding a main effect of audio model, the experimenters continued to do a follow up analysis of contrasts to correctly identify which models performed the best for each factor. In accuracy, AudioGen performed the best with a contrast estimate value of 2.272. In fidelity, CTAG performed the best with a contrast estimate value of 1.311. And lastly, in Artistic Interpretation, AudioGen performed the best with a contrast estimate value of 1.483. On the other end, AudioLDM performed the worst in accuracy and fidelity and CTAG performed the worst in Artistic Interpretation. These findings align with the data collected and displayed in the line graphs in Figure 3, 4, and 5.

3.3 Comparative Analysis

This finding was quite similar to what Cherep et al. had found in their user study with the same three models. AudioGen had performed the best and AudioLDM had performed the worst with CTAG being somewhere in the middle. This study differed in that the experimenters chose to rate fidelity instead of confidence and all three metrics were rated on 7 point Likert scales rather than accuracy being rated as a percentage. While the results were quite similar, the biggest difference was that CTAG was rated as statistically significantly higher than the other two models in the category of fidelity, which was not a variable tested in the paper by Cherep et al.

3.4 Qualitative Results

Qualitative analysis identified three key trends from analyzing the comments from the participants.

3.4.1 Perceived Intuitiveness of UI

Participant 11 described UI2 as "intuitive and professional," with many citing the advanced features and customizable controls as enhancing their evaluation process. In contrast, UI1 was labeled as "bland" by participant 2 and "simplistic" by participant 17, with some users reporting difficulty in making detailed assessments even though the audios presented with both GUIs were identical.

3.4.2 Influence of Visual Context

The presence of detailed a GUI influenced qualitative comments with several participants noting that UI2 made the audios "sound more spatial" (participant 1) and "have more dynamic range" (participant 6) to the audio files, even when the audio itself was unchanged.

3.4.3 Preferences for AI Generators

AudioGen was frequently praised for its "realistic" output, while AudioLDM was critiqued for sounding "mechanical" or "indistinct." CTAG was preferred overall in the fidelity category but had less qualitative feedback than the other text-to-audio generators. Some participants appreciated the study's structure and some even wrote in compliments in the final comments section. One participant reported that the audios were the same for both GUI designs.

4 DISCUSSION

How human-computer interaction impacts the perception of AI generated sound tools is of central interest to the development of modern technology that facilitates creativity and expression. While the study did not reveal and statistically significant differences between two GUIs, qualitative responses revealed that small differences in the GUIs changed participants' perception of audio and can be further studied with more distinct differences between GUIs.

4.1 Limitations

The study recruited 21 participants exclusively from the Frost School of Music at the University of Miami, all of whom were music engineering students. Additionally, The study only used a single text prompt to generate audio samples. The experimenters focused on three AI audio generation models: AudioGen, CTAG, and AudioLDM. The GUIs were very similar and only had 5 different elements between them, with font, color, text size, and overall look and feel being exactly the same. Participants completed the survey on their personal devices in varied environments. All of these limitations can be expanded upon with more resources and further attention to detail.

4.2 Future Work

This study highlights several promising avenues for future research into AI-generated audio and its interaction with user interfaces. First, expanding the participant pool to include individuals from diverse backgrounds and expertise levels could provide broader insights into how dif-

ferent user groups perceive AI-generated audio. Including non-expert users, hobbyists, and individuals from varied cultural contexts would help generalize findings and uncover unique preferences or biases. Additionally, testing a wider range of text prompts, genres, and scenarios, such as sound effects, musical compositions, and environmental audio, could offer a more comprehensive understanding of how AI audio systems perform across diverse creative applications. Incorporating prompts with varying complexity or abstractness could also shed light on how users interpret and evaluate AI-generated outputs under different conditions.

Another important direction involves further refining the design of graphical user interfaces (GUIs) to explore how distinct design elements influence user perception and engagement. Future studies could introduce more dramatic variations in interface aesthetics, functionality, and interactivity to identify which design features have the most significant impact on user experience. For example, including dynamic visualizations, real-time audio manipulation tools, or more personalized customization options could enhance user interaction and creativity. Moreover, integrating multimodal stimuli, such as pairing audio with visual or textual content, could uncover new dimensions in the perception of AI-generated outputs.

5 REFERENCES

- [1] H. Zhang, F. Yang, "HCI: A journey through sound," *Universal Access in the Information Society*, vol. 20, no. 1, pp. 1–15 (2021), URL <https://doi.org/10.1007/s00779-021-01554-z>.
- [2] X. Li, J. McHugh, "Human-AI interaction in music," presented at the *ISMIR* (2023), URL <https://archives.ismir.net/ismir2023/paper/000008.pdf>.
- [3] A. Agostinelli, T. I. Denk, Z. Borsos, J. Engel, M. Verzett, A. Caillon, Q. Huang, A. Jansen, A. Roberts, M. Tagliasacchi, M. Sharifi, N. Zeghidour, C. Frank, "MusicLM: Generating Music From Text," *arXiv*, no. arXiv:2301.11325 (2023), URL <http://arxiv.org/abs/2301.11325>.
- [4] J. Copet, F. Kreuk, I. Gat, T. Remez, D. Kant, G. Synnaeve, Y. Adi, A. Défossez, "Simple and Controllable Music Generation," *arXiv*, no. arXiv:2306.05284 (2024), URL <http://arxiv.org/abs/2306.05284>.
- [5] J. Kwon, J. Lee, "User-centered design of AI music generation systems: A framework for enhancing user experience," *International Journal of Human-Computer Studies*, vol. 171, p. 102806 (2023), URL <https://doi.org/10.1016/j.ijhcs.2023.102806>.
- [6] M. Cherep, N. Singh, J. Shand, "Creative text-to-audio generation via synthesizer programming," *arXiv* (2024), URL <https://arxiv.org/abs/2406.00294>.
- [7] D. Ghosal, N. Majumder, A. Mehrish, S. Poria, "Text-to-audio generation using instruction-guided latent diffusion model," *arXiv* (2024), URL <https://arxiv.org/abs/2406.00294>.
- [8] R. Ranjan, R. Tiwari, M. Kumar, "A systematic review of artificial intelligence-based music generation: Scope, applications, and future trends," *Journal of Cleaner Production*, vol. 352, p. 131195 (2022), URL <https://doi.org/10.1016/j.jclepro.2021.131195>.
- [9] E. R. Miranda, "An artificial intelligence approach to sound design," *Computer Music Journal*, vol. 19, no. 2, pp. 59–75 (1995), doi:10.2307/3680600, URL <https://doi.org/10.2307/3680600>.
- [10] H. Browne, A. Tursunov, "Generative AI and HCI," *Proceedings of the ACM on Human-Computer Interaction*, vol. 6, no. CSCW2, p. Article 408 (2022), URL <https://doi.org/10.1145/3491101.3503719>.
- [11] T. Falk, H. Chen, "Music and AI: A relationship based on creativity and collaboration," *AI Society*, vol. 36, no. 3, pp. 839–851 (2021), URL <https://doi.org/10.1007/s00146-020-01030-1>.
- [12] X. Huang, L. Wang, "Collaborative AI for music creation: Enhancing the creative process through intelligent interfaces," *ACM Transactions on Interactive Intelligent Systems*, vol. 13, no. 2, pp. 1–25 (2023), URL <https://doi.org/10.1145/3455079>.
- [13] S. Leroy, A. Py, "Human-AI musical computing framework," (2018), URL <https://hal.science/hal-01791504/>.
- [14] Y. Wang, Y. Chen, "Exploring the impact of AI on music composition: A user study of AI-assisted music creation tools," *Journal of Music Technology and Education*, vol. 15, no. 1, pp. 45–60 (2022), doi:10.1386/jmte-00006-1, URL <https://doi.org/10.1386/jmte-00006-1>.
- [15] S. Hwang, "Investigating the physiological and psychological effect of an interactive musical interface for stress and anxiety reduction," *MIT Media Lab* (2020), URL <https://www.media.mit.edu/publications/investigating-the-physiological-and-psychological-effect-of-an-interactive-musical-interface-for-stress-and-anxiety-reduction/>.
- [16] S. Shah, R. Trivedi, "Dialogue HCI and musical interaction," presented at the *Proceedings of the ACM on Human-Computer Interaction*, vol. 7, p. Article 11 (2023), URL <https://doi.org/10.1145/3613904.3642171>.
- [17] D. Thompson, R. Carter, "Music creation by example," presented at the *Proceedings of the ACM on Human-Computer Interaction*, vol. 2, p. Article 204 (2018), URL <https://doi.org/10.1145/3313831.3376514>.
- [18] L. Yang, "Novice AI in music creation," presented at the *Proceedings of the ACM on Human-Computer Interaction*, vol. 2, p. Article 205 (2018), URL <https://doi.org/10.1145/3313831.3376739>.
- [19] A. Miller, "How people perceive AI-generated music," presented at the *Proceedings of the ACM on Human-Computer Interaction*, vol. 7, p. Article 58 (2023), URL <https://doi.org/10.1145/3511808.3557235>.

[20] M. Newman, L. Morris, J. H. Lee, “Human-AI music creation: Understanding the perceptions and experiences of music creators for ethical and productive collaboration,” presented at the *Proceedings of the International Society for Music Information Retrieval Conference (ISMIR)* (2023), URL <https://archives.ismir.net/ismir2023/paper/000008.pdf>.